

The empirical recurrence for OEIS A262920: recovering a conjecture that is not written down

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Abstract

OEIS A262920 records “Empirical recurrence of order 65”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 360$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 65, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 65 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A262920 is “Number of $(6+1)X(n+1)$ 0..1 arrays with each row divisible by 3 and each column divisible by 7, read as a binary number with top and left being the most significant bits.”. It has offset 1 and begins

19, 61, 1207, 7417, 152587, 1550557, 30497815, 420921961, ...

The entry states:

Empirical recurrence of order 65 (see link above)

As of the “Last modified” line on the live entry (Oct 04 2015 10:16:19, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2187$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 360$ of the 2187, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 360$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 720$ exact terms of the model returns order 65 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{65} c_i a(n-i),$$

c_1	19	c_{18}	3782061958750296465489582	c_{35}	−424067037495
c_2	2819	c_{19}	−71859177216255632844302058	c_{36}	−104488073696
c_3	−53561	c_{20}	−615753950181021182769933858	c_{37}	1985273400233
c_4	−3758893	c_{21}	11699325053439402472628743302	c_{38}	4178032961498
c_5	71418967	c_{22}	84944258101550579450681793990	c_{39}	−7938262626846
c_6	3157693687	c_{23}	−1613940903929461009562954085810	c_{40}	−14206889727069
c_7	−59996180053	c_{24}	−9985084342989824039953741160430	c_{41}	269930904814328
c_8	−1877872245671	c_{25}	189716602516806656759121082048170	c_{42}	408431157283465
c_9	35679572667749	c_{26}	1004136326485405859516468650728810	c_{43}	−776019198838584
c_{10}	842192393920261	c_{27}	−19078590203222711330812904363847390	c_{44}	−985263679084078
c_{11}	−16001655484484959	c_{28}	−86614086501662908120826112123081510	c_{45}	18720009902597496
c_{12}	−296284641226998003	c_{29}	1645667643531595254295696130338548690	c_{46}	1975142947162002
c_{13}	5629408183312962057	c_{30}	6417121216724171593889318827309193490	c_{47}	−3752771599607804
c_{14}	83965694546993923881	c_{31}	−121925303117759260283897057718874676310	c_{48}	−3249964011081422
c_{15}	−1595348196392884553739	c_{32}	−408468513152175087076632060787142834085	c_{49}	617493162105470209
c_{16}	−19533415046160113029674	c_{33}	7760901749891326654456009154955713847615	c_{50}	431969339262346599
c_{17}	371134885877042147563806	c_{34}	22319317762913700299575743970961851487055	c_{51}	−82074174459845852

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 65, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $65 + 4$ terms removed returns order 65 as well, so $\deg q_{\min} = 65 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $65 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 65 that this sequence satisfies — and that family turns out to have exactly one member.

References

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